Module 11

Implementing Performance and Monitoring

Module Objectives

- Examine tools and strategies for optimizing various cloud services.
- Review essential scaling techniques.
- Examine observability options and configurations.
- Understand the lifecycle of cloud resources from development and deployment through maintenance and decommissioning.

Lesson 11.1

Optimize Workloads Using Cloud Resources

Cloud+ CV0-004 Certification Exam Objective:

- 1.10 Compare and contrast methods for optimizing workloads using cloud resources.

Optimize Workloads Using Cloud Resources

- Implement Compute Resources
- Compute Resources for Virtual Machines
- Compute Resources for Containers
- Compute Resources for Serverless Applications
- Storage Workloads
- Network Traffic
- Network Workloads
- Orchestration and Workflow Optimization
- Managed Services

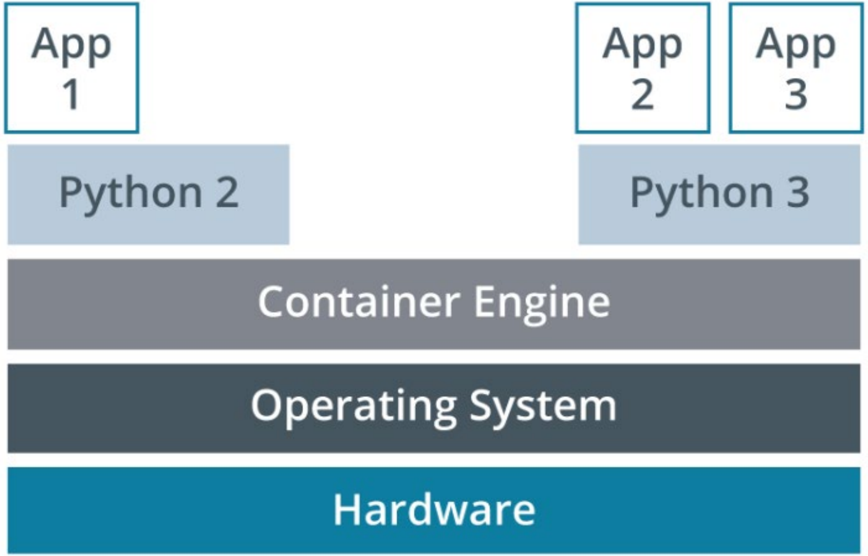
Implement Compute Resources

- Components
- VM instance types
- Container optimization
- Serverless computing

Compute Resources for Virtual Machines

- CPU considerations
- GPU considerations
- Memory considerations

Compute Resources for Containers



App1 requires a Python 2 container, and Apps 2 and 3 require a Python 3 container.

Compute Resources for Serverless Applications

- Characteristics
- Optimization
- Monitoring tools
    - AWS
    - Azure
    - GCP

Storage Workloads

- Storage tiers
- Storage input-output per second (IOPS)
- Storage throughput
- Storage capacity
- Data deduplication
- Data compression

Network Traffic

- Network bandwidth
- Network throughput
- Network latency

Network Workloads

- Software-defined network (SDN)
- Content delivery networks (CDN)
- Incorrectly configured or failed load balancing

Orchestration and Workflow Optimization

- Orchestration optimization
- Workflow optimization

Managed Services

- Architecture
- Operations
- Support
- Hosting

Lesson 11.2

Configure Appropriate Scaling Approaches

Cloud+ CV0-004 Certification Exam Objective:

- 3.2 Given a scenario, configure appropriate scaling approaches.

Configure Appropriate Scaling Approaches

- Scaling Approaches
- Configuration Types for Scaling
- Troubleshoot Auto-scaling Issues

Scaling Approaches

- Manual vs. scheduled scaling
- Triggered scaling
  - Trending
  - Load
  - Event

Configuration Types for Scaling

- Horizontal scaling
- Vertical scaling
- Cloud bursting

Troubleshoot Auto-scaling Issues

- Scaling is not achieving the desired results
  - Autoscaling is configured
  - Scaling out is a benefit for the application or service
  - Scaling up is a benefit for the application or service

Lesson 11.3

Configure Resources to Achieve Observability

Cloud+ CV0-004 Certification Exam Objective:

- 3.1 Given a scenario, configure appropriate resources to achieve observability.

Configure Resources to Achieve Observability

- Cloud Service Observability
- Logging Collection
- Log Aggregation
- Log File Analysis
- Log File Retention
- Tracing
- Optimization and Monitoring
- Cloud Detection and Response Solutions

Cloud Service Observability

- Tagging
- Costs
- Elasticity usage
- Connectivity
- Latency
- Incidents
- Overall utilization

Logging Collection

- Rsyslog
- Event Viewer

Log Aggregation

- Linux servers
- Windows servers
- Cloud logging

Log File Analysis

Information	3/24/2021 7:47:52 AM
Error	3/24/2021 7:47:53 AM
Information	3/24/2021 7:47:53 AM
Information	3/24/2021 7:47:53 AM
Information	3/24/2021 7:47:53 AM
Information	3/24/2021 7:47:52 AM
Warning	3/24/2021 7:47:52 AM

Event Viewer severity levels. (Screenshot courtesy of Microsoft.)

Kernel constant	Level value	Meaning
KERN_EMERG	0	System is unusable
KERN_ALERT	1	Action must be taken immediately
KERN_CRIT	2	Critical conditions
KERN_ERR	3	Error conditions
KERN_WARNING	4	Warning conditions
KERN_NOTICE	5	Normal but significant condition
KERN_INFO	6	Informational
KERN_DEBUG	7	Debug-level messages

rsyslog severity levels.

Log File Retention

- Audit logs
- Types of logs
  - Authentication logs
  - File access logs
  - System logs
  - Application logs
- Retention benefits
- Log scrubbing

Tracing

- Records each function call within the code or application
- Allows troubleshooting and optimization by providing detailed information of the application's processing steps
- Tools
  - Google Cloud Trace
  - Azure Monitor
  - AWS X-Ray

Optimization and Monitoring

- Metrics
- Baselines
- Thresholds, performance monitoring, and resource utilization
- Integrated service management tools

Cloud Detection and Response Solutions

- Incident alerts
- Incident responses
- Enable and disable alerts

Lesson 11.4

Managing the Life Cycle of Cloud Resources

Cloud+ CV0-004 Certification Exam Objective:

- 3.4 Given a scenario, manage the life cycle of cloud resources.

Managing the Life Cycle of Cloud Resources

- The Lifecycle Roadmap
- Major and Minor Updates
- Patches
- Update and Patch Testing
- Ephemeral and Persistent Data

The Lifecycle Roadmap

- Development phase
- Deployment phase
- Maintenance phase
- Deprecation phase
- Decommissioning phase

Major and Minor Updates

- Versioning System:
  - Semantic versioning: major.minor.patch
- Major updates
- Minor updates

Patches

- Reasons for patching
- Scope of cloud elements to be patched
- Patch management policies

Update and Patch Testing

- Ensures functionality and compatibility of updates and patches
- Testing environments
- Deployment strategies
  - Blue-green deployment
  - Phased rollout

Ephemeral and Persistent Data

Persistent Data

- Resides on long-term storage devices (e.g., SSDs, HDDs)
- Data Management
  - Physical security
  - Encryption
  - Permissions
  - Backups
  - Replication

Ephemeral Data

- Exists in volatile storage (e.g., RAM)
- Data Management
  - Minimal management needed as data is discarded after use

Module Summary

- Observability: View data to improve resource utilization and customer experience.
- Optimization: Use third-party managed services, effective automation, and efficient processes.
- Resource Scaling: Ensure availability and access.
- Benefits: Ensures excellent performance and controls costs.